

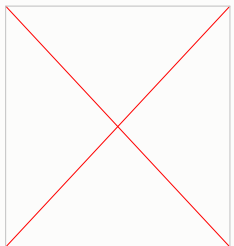
MATSim User Meeting 2026 · Berlin

Agent-Based Modeling of a Zero-Emission Zone in Berlin Using MATSim

Fleet electrification, adaptation dynamics, and assessment of emission impacts

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Policy: Zero-Emission Zone inside the S-Bahn Ring

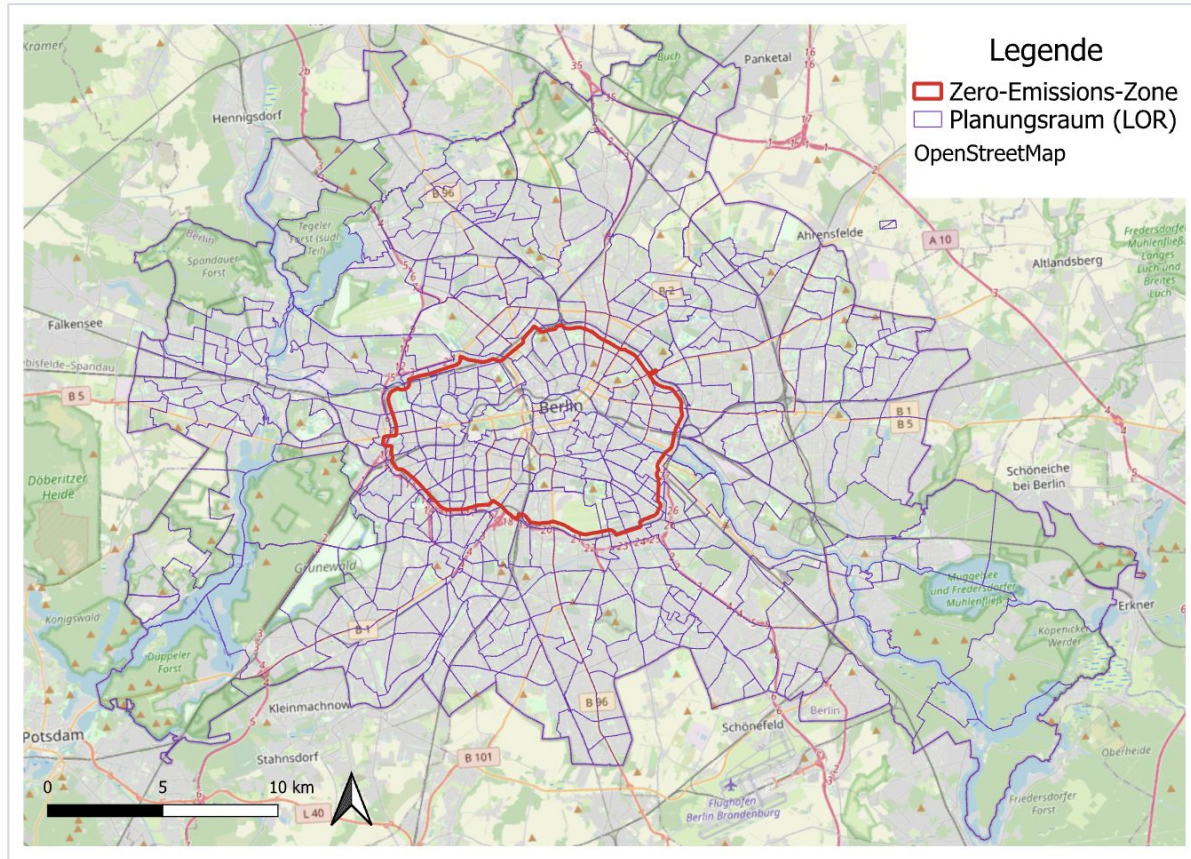


Figure 1: ZEZ boundary (red) and LOR planning areas. Map data: OpenStreetMap.

- All road links inside the S-Bahn Ring, plus a buffer of about 500 m, are closed to combustion-engine vehicles
- Electric vehicles of all categories keep unrestricted access
- Objectives: lower CO₂, NO_x and PM emissions and external costs (health, climate)

Research questions

1. EV adoption grows regardless of additional policy. How much does a ZEZ accelerate this transition?
2. To what extent does this acceleration produce environmental and societal benefits?
3. Which negative effects may arise, in particular for different income groups?

Methodological contributions

1 Long-term EV use in MATSim

EV modes with their own cost structure; relaxed long-term equilibrium reached by initialising full-EV plans.

2 Diffusion curve

Links the MATSim equilibrium to the observed pace of fleet renewal, with separate curves for residents inside and outside the zone.

3 MATSim scoring as the welfare measure

The only consistent way to account for each agent's gains and losses across all adaptation mechanisms.

Modeling framework

1 • Transport

MATSim: daily trips, modes and routes. Impact = Δ between policy and reference.



2 • Emissions

Emissions contrib with HBEFA factors per link traversal: CO₂, NO_x, PM.



3 • Monetization

MATSim score for users, externalities and implementation costs in a CBA.



4 • Distribution

Welfare changes by income decile.

Model: open Berlin MATSim scenario v6.4

- Base year 2023; public transport schedule updated to 2024
- Activity-based daily plans for the Berlin metropolitan region
- Modes: car, ride, pt, bike, walk; car + ride = MIT for comparison with MiD
- Commercial and freight traffic as separate agent groups
- Evaluation in 2022 prices
- ZEZ implemented on the network: links inside the zone removed for car
- Replanning: SubtourModeChoice, re-routing
- Output: modal split, vehicle-km and speeds per link as input for HBEFA and CBA

Scenario design

Stage	Reference (no ZEZ)	Policy (ZEZ)
Baseline 2023	Baseline	—
Relaxed: long-term equilibrium, no annual fleet constraint	EV-relaxed	EV&ZEZ-relaxed
2035: relaxed results scaled by fleet diffusion	EV'35	EV&ZEZ'35

Policy impact: Δ = policy – reference, evaluated both in the relaxed state and for 2035.

Long-term EV use in MATSim

- Additional modes `ev_car` and `ev_ride` in `SubtourModeChoice`
- EVs: higher fixed daily cost, lower cost per km
- Relaxed equilibrium = long-term state under given prices and rules, without an annual fleet constraint
- In the ZEZ scenario, car links inside the zone are unavailable; `ev_car` keeps full access

Cost component	car	ev_car
Fixed cost per day	1.0×	1.1×
Variable cost per km	1.0×	0.8×

Table 1: relative cost parameters (car = 1)

car / ev_car local equilibrium

Problem

Most agents start with car plans. With a high fixed daily cost, switching single subtours to ev_car rarely improves the score. Replanning converges to a local equilibrium with a strongly underestimated EV share.

Result: about **+25%** agents using electric vehicles in the final equilibrium.

Solution

For each relevant agent, add one plan identical to the original but with all car legs replaced by ev_car. Its initial score is null, so MATSim evaluates it in the first iterations.

From equilibrium to fleet dynamics

MATSim

Long-term behavioural equilibrium under given conditions: the target EV share K_{EV} .

Fleet renewal

Actual fleet renewal is gradual: new registrations follow an S-curve; average vehicle lifetime T .

$$q(t) = K_{EV} / (1 + \exp(-r \cdot (t - t_0)))$$

Logistic diffusion of EV new registrations; saturation level taken from the MATSim relaxed run

Calibration on Berlin registrations

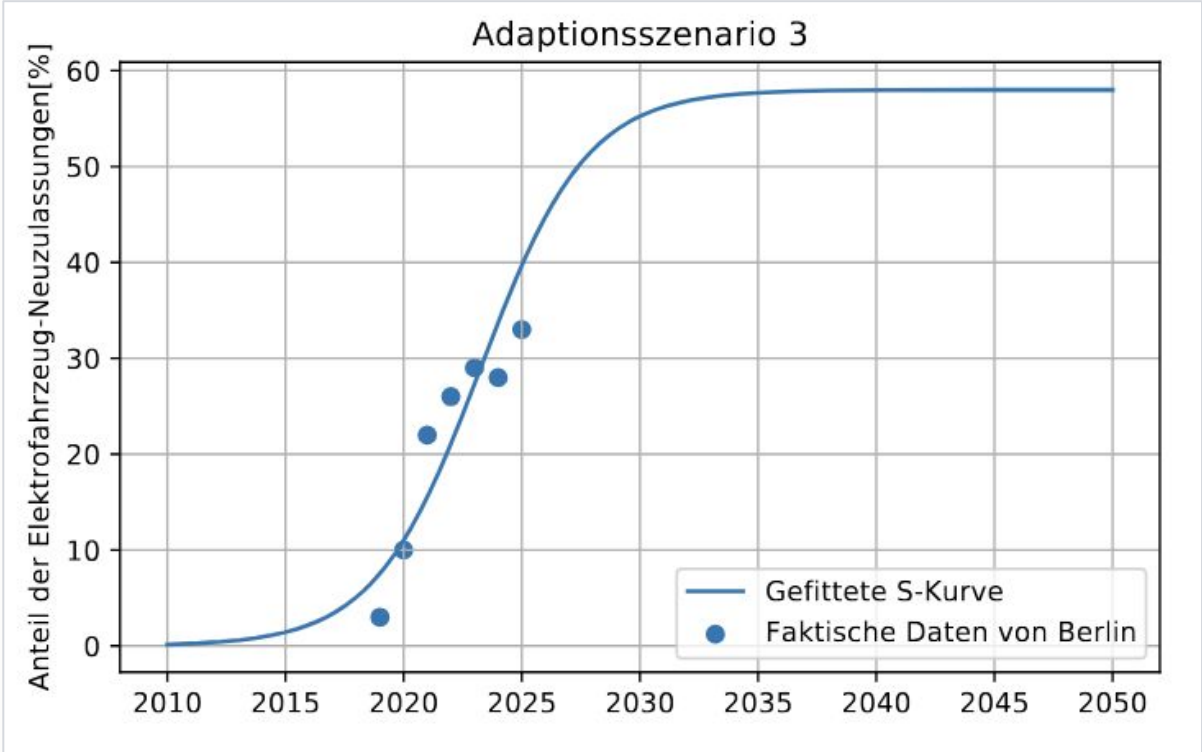


Figure 2: EV share of new registrations in Berlin, fitted S-curve vs. observed data.

Parameter	Value
K_EV (EV-relaxed)	0.58
r	0.4
t ₀	2023
Vehicle lifetime T	13 years
New registrations K	80,000 / year

Curves inside vs. outside the zone

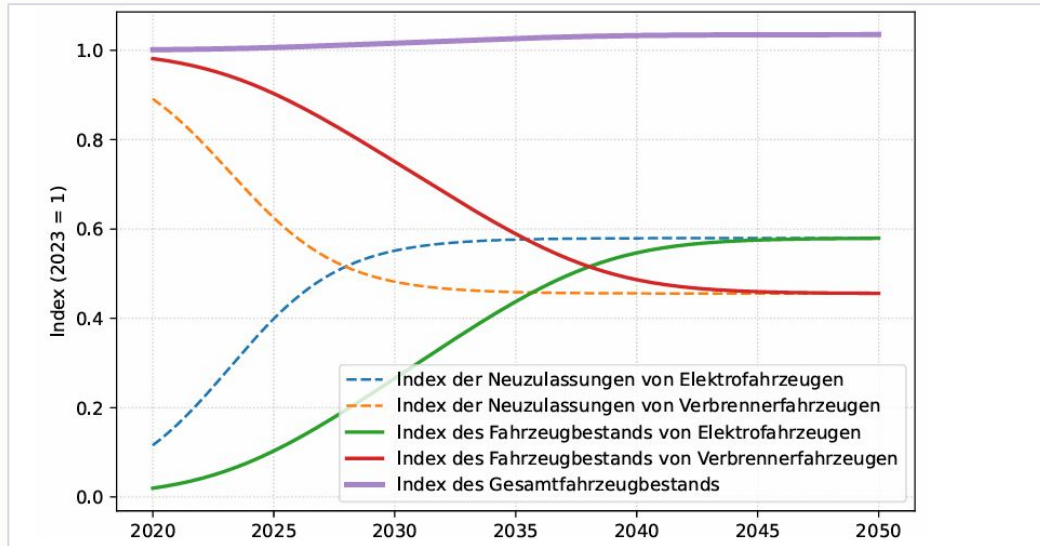


Figure 3a: residents outside the ZEZ, T = 13 years.

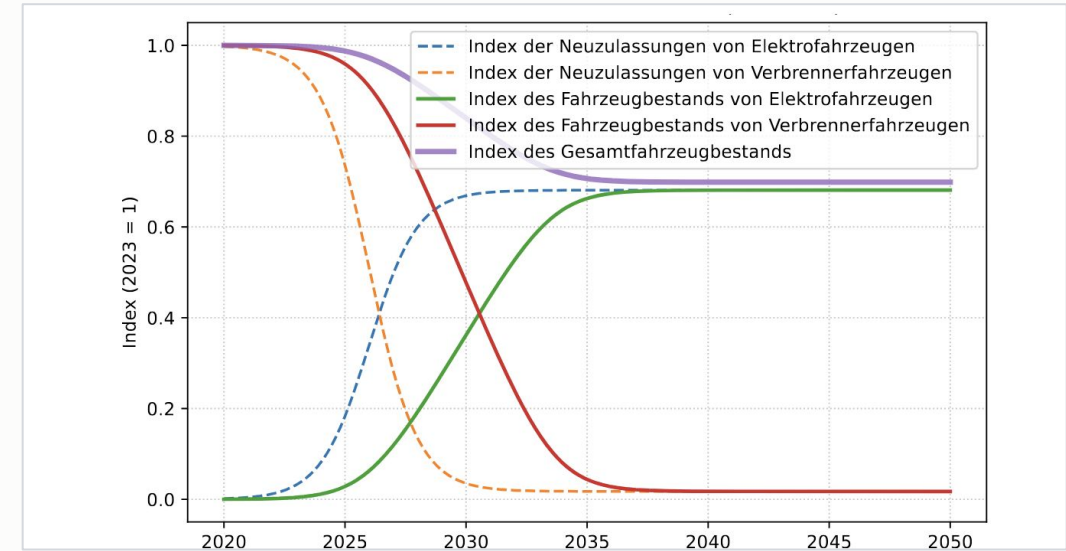


Figure 3b: residents inside the ZEZ, T = 6.5 years.

- Inside the zone, fleet renewal is accelerated: owner replacement and redistribution via the second-hand market
- 2035 EV share: both trajectories aggregated, weighted by the trip volumes of each group

Progression factor for 2035

Scenario	Group	I_{2035}	I_{end}	p_{2035}
EV-relaxed	Berlin	0.41	0.58	70.7%
EV&ZEZ-relaxed	Berlin outside the ZEZ	0.48	0.63	76.4%
EV&ZEZ-relaxed	Trips to, from, within the ZEZ	–	–	100%

$p_{2035} = I_{2035} / I_{end}$: share of the equilibrium EV shift realised by 2035. Trips touching the zone reach the equilibrium state.

MATSim score as welfare measure

$$S_{plan} = \sum S_{act,q} + \sum S_{trav,mode}(q)$$

Activity utility (+) and travel disutility (–: time, detours, transfers, monetary cost)

Why the score

Agents adapt in several ways at once: EV purchase, mode shift to PT or bike, re-routing, changed trip chains. All of them change one consistent per-agent utility.

Use in the assessment

Δ score (policy – reference), converted to € via the marginal utility of money, gives each agent's gain or loss: input to the CBA and to the income-group analysis.

Cost-benefit analysis 2035

Component	Source
Private users	MATSim scoring
Commercial transport	Financial scoring
CO ₂ , NO _x , PM	HBEFA + UBA cost rates
Noise	German EWS guidelines
Accidents	Statistical analysis
Infrastructure, administration	Literature review

Pollutant	€ / t
CO ₂ (0% time preference)	901
NO _x (transport, city)	38,808
PM ₁₀	60,799
HC	554
SO ₂	36,221

Table 2: cost rates, UBA Methodological Convention 3.2

Results: modal split in the relaxed equilibrium

Mode (Berlin residents)	EV-relaxed	EV&ZEZ-relaxed	Δ pp
Walk	29.62%	29.80%	+0.18
Public transport	28.05%	29.64%	+1.59
Bike	17.65%	18.54%	+0.89
MIT	24.68%	22.02%	-2.66
– fossil	11.33%	6.35%	-4.98
– electric	13.35%	15.67%	+2.32

ZEZ residents

- MIT: 16.05% → 11.21%
- Combustion car: 0.27%
- Electric MIT: 10.95%
- PT: 36.61%, bike: 23.60%

Table 3: modal split. Baseline MIT share: 23.84% (rebound effect in EV-relaxed).

Results: modal split in 2035

Car trips, share of all trips	EV'35	EV&ZEZ'35	Δ pp
MIT	24.68%	22.02%	-2.66
– fossil	15.43%	9.30%	-6.13
– electric	9.25%	12.72%	+3.47

- Diffusion redistributes car trips between fossil and electric; walk, PT and bike shares are taken from the relaxed runs
- Among ZEZ residents almost all remaining car traffic is electric

Results: emissions 2035 (EV&ZEZ'35 vs. EV'35)

Pollutant / vehicle class	Δ t/day
CO ₂ , total	-2,413
– passenger cars	-1,033
– light commercial vehicles	-855
– trucks	-524
NO _x	-5.89
CO	-19.72
HC	-0.90
PM	-0.07

Table 4: direct emissions, Berlin. Outside the zone alone: -234 t CO₂/day.

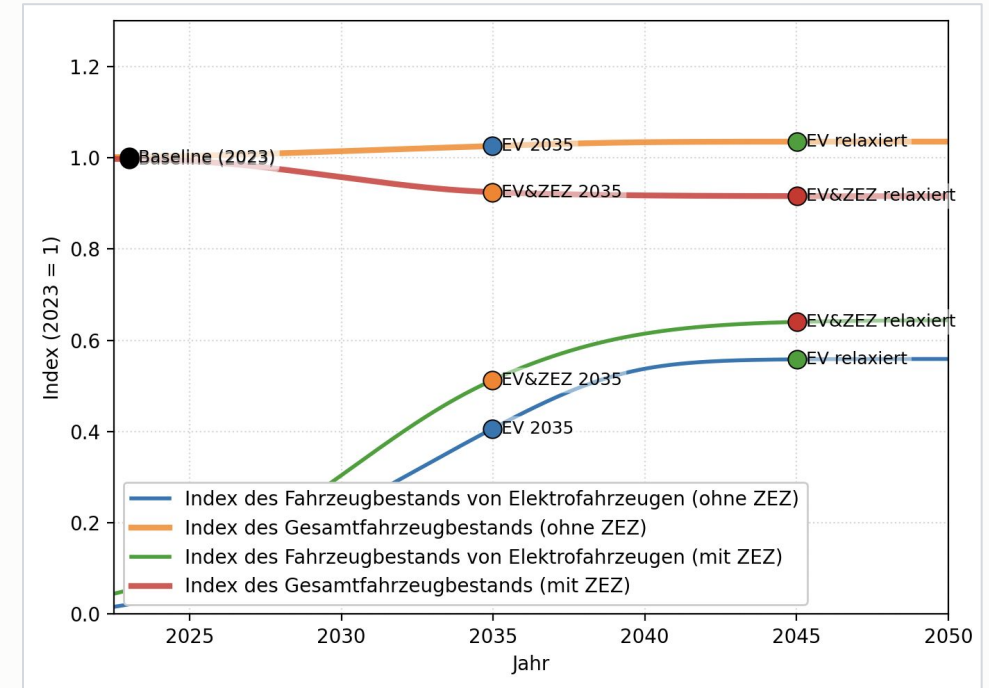


Figure 4: emission index (NO_x, CO₂; 2023 = 1) without and with ZEZ.

Results: noise and accidents 2035

Impact, Berlin	$\Delta \text{ € / day}$
Noise damages	-44,000
Accident costs	-171,000

Table 5: negative values = reduction

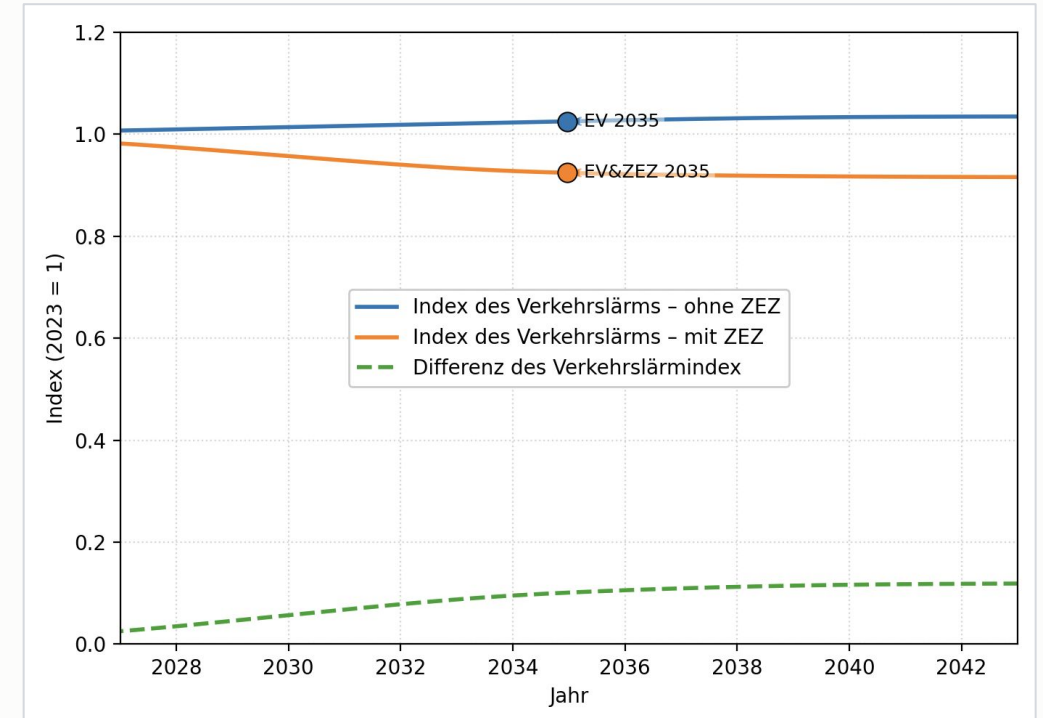


Figure 5: traffic noise index (2023 = 1) without and with ZEZ.

Results: welfare change by income group

Population group	Δ welfare per person (relative)
All agents	-1×
Top 10% income	-5×
Bottom 10% income	≈ 0

Table 6: change in MATSim score per person and day, relative to the population average

- The top income decile carries the largest losses: these agents are the heaviest car users
- Since the score is computed per agent, the distributional analysis requires no additional model

Conclusions

Findings

- The ZEZ accelerates fleet electrification
- It reduces MIT in the centre, shifting trips to PT and cycling
- Additional reduction of about 2,400 t CO₂ per day in 2035
- Net effect is positive but sensitive to CO₂ cost assumptions

Method and outlook

- MATSim equilibrium + diffusion model capture both behavioural adaptation and the pace of fleet renewal
- MATSim score as a consistent welfare measure per agent
- Next: home-charging access as a determinant of variable EV costs

Thank you for your attention

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